

Healthcare Business Capstone

Hospital Operations & Revenue Risk Intelligence Platform

1. Business Introduction

You are part of the AI Transformation and Analytics Team for a multi-specialty hospital network operating across multiple cities. The hospital leadership is facing challenges in managing patient flow, controlling operational inefficiencies, and minimizing financial losses due to insurance claim rejections and delayed payments.

The hospital has recently consolidated its data systems into three core operational datasets and wants to leverage SQL, Python, and Artificial Intelligence to build a decision-support platform that can assist administrators, doctors, and finance teams in making faster, more reliable, and more profitable decisions.

Your role is not to build an academic model. Your role is to design and deploy a business-grade analytics and AI system that could realistically be proposed to hospital leadership as a deployable solution.

2. Capstone Objectives

By the end of this capstone, learners are expected to achieve the following business and technical objectives:

- Design a structured SQL-based analytics layer for hospital operations and financial data.
- Perform comprehensive Exploratory Data Analysis to identify trends, risks, and inefficiencies.
- Build classification models to predict patient visit risk and insurance claim outcomes.
- Evaluate models using both technical performance metrics and business impact metrics.
- Deploy trained models as an API service that can be used by dashboards or internal hospital systems.
- Implement basic monitoring, data validation, and governance mechanisms for production readiness.
- Communicate insights and system value through a business-focused executive report.

3. Data Description

You are provided with three CSV files that simulate real-world hospital operational and financial systems. These files represent anonymized and synthetic healthcare data designed

to support SQL analysis, data science workflows, and machine learning model development.

You must load these CSV files into a relational SQL database of your choice (for example, MySQL, PostgreSQL, or SQL Server) and design your analytics and modeling workflows on top of the SQL layer.

You are responsible for creating appropriate primary keys, foreign key relationships, indexes, and data validation rules.

3.1 Dataset Overview

File Name	Table Name (SQL)	Purpose	Key Relationships
patients.csv	patients	Stores patient demographic and registration data	Primary key referenced by visits
visits.csv	visits	Stores hospital encounter and operational data	References patients, referenced by billing
billing.csv	billing	Stores billing and insurance claim information	References visits

3.2 Patients Table Description

Column Name	Data Type (Suggested)	Business Meaning
patient_id	INTEGER	Unique identifier for each patient
age	INTEGER	Patient age, used as a clinical risk factor
gender	VARCHAR	Patient gender for demographic analysis
city	VARCHAR	Patient location for regional demand analysis
insurance_provider	VARCHAR	Insurance company covering the patient
chronic_flag	BOOLEAN / INTEGER	Indicates long-term medical condition status
registration_date	DATE	Date when patient first registered with the hospital

3.3 Visits Table Description

Column Name	Data Type (Suggested)	Business Meaning
visit_id	INTEGER	Unique identifier for each hospital visit
patient_id	INTEGER	Links visit to patient record
visit_date	DATE	Date of hospital visit

department	VARCHAR	Department where patient was treated
visit_type	VARCHAR	Type of visit such as OPD, ER, or ICU
length_of_stay_hours	DECIMAL	Duration of stay used as an efficiency metric
risk_score	VARCHAR	Risk classification for patient visit
doctor_id	INTEGER	Identifier for attending doctor

3.4 Billing Table Description

Column Name	Data Type (Suggested)	Business Meaning
bill_id	INTEGER	Unique billing transaction identifier
visit_id	INTEGER	Links billing record to hospital visit
billed_amount	DECIMAL	Total amount billed by hospital
approved_amount	DECIMAL	Amount approved by insurance provider
claim_status	VARCHAR	Final status of insurance claim
payment_days	INTEGER	Number of days taken to receive payment
billing_date	DATE	Date when bill was generated

4. SQL Loading Instructions

Learners must create a relational database and load the CSV files into SQL tables before beginning analysis and modeling.

Minimum requirements:

1. Create tables using appropriate data types and constraints.
2. Define primary keys for each table.
3. Define foreign key relationships between patients, visits, and billing tables.
4. Create indexes on commonly queried fields such as visit_date, department, insurance_provider, and claim_status.

All further analysis, feature engineering, and model-building workflows must be driven from SQL queries or SQL-extracted datasets.

Phase 1 — SQL Analytics Layer (Business Intelligence Foundation)

Business Goal

Create a reliable, queryable hospital data layer that leadership can trust for operational and financial decision-making.

Technical Objectives

Transform raw CSV files into a structured relational database, enforce data integrity, and create reusable business views for analysis and reporting.

Tasks

- Create a new database schema for the hospital system.
- Define SQL tables: patients, visits, and billing with appropriate data types.
- Define primary keys and foreign key relationships between tables.
- Document which queries benefit from which indexes.
- **Operational Analysis**
 - Retrieve the **top 10 departments by total visit volume**.
 - Identify the **top 5 departments with the highest average length of stay**.
 - Find the **percentage of High Risk visits per department**.
 - Determine the **average number of visits per patient by city**.
 - Identify **doctors handling the highest number of High Risk visits**.
- **Financial Analysis**
 - Retrieve the **top 10 insurance providers by total billed amount**.
 - Identify the **top 5 insurance providers with the highest claim rejection rate**.
 - Find the **average payment delay (payment_days) by insurance provider**.
 - Calculate the **revenue realization ratio (approved_amount / billed_amount) by department**.
 - Identify **visits where billed_amount is high but approved_amount is zero or missing**.
- **Data Quality and Integrity Checks**
 - Detect **visits that do not have a corresponding billing record**.
 - Detect **billing records that do not have a corresponding visit**.
 - Identify **patients with duplicate patient_id values**.
 - Find records with **missing or invalid length_of_stay_hours or payment_days**.
 - Identify **visits linked to patients with missing insurance provider information**.

Phase 2 — Exploratory Data Analysis and Data Quality

Business Goal

Understand hospital operations, financial performance, and data reliability before deploying AI models.

Tasks

- Load data into Python for analysis. Combine them
- Analyze missing values in `approved_amount`, `payment_days`, and `length_of_stay_hours`.
- Perform distribution analysis by department, visit_type, insurance_provider, and city.
- Detect and classify outliers in `billed_amount`, `payment_days`, and `length_of_stay_hours`.
- Engineer features such as visit frequency, average length of stay per patient, provider rejection rate, days since registration, and time-based features.

Deliverables

- EDA notebook (01_eda.ipynb)
- Feature engineering script (build_features.py)
- Modeling dataset (model_table.csv or parquet)
- Data quality report (docx or markdown)

Phase 3 — Model Development (Classification Systems)

Model A — Visit Risk Classification

Business Purpose: Predict whether a hospital visit represents a Low, Medium, or High operational and clinical risk.

- Define the target variable as `risk_score`.
- Select and justify feature set based on business relevance.
- Perform a time-based train and test split (earliest 80 percent for training, latest 20 percent for testing).
- Train a baseline model using Logistic Regression.
- Train an advanced model such as Random Forest or Gradient Boosting.
- Perform optional hyperparameter tuning and document results.

Model B — Claim Outcome Classification

Business Purpose: Predict whether an insurance claim will be Paid, Pending, or Rejected before submission.

- Define the target variable as `claim_status`.
- Engineer financial and operational predictor features.
- Perform a time-based train and test split.
- Train baseline and advanced classification models.
- Analyze class imbalance and mitigation strategy.

Deliverables

- Risk model notebook (02_risk_model.ipynb)
- Claim model notebook (03_claim_model.ipynb)
- Saved model artifacts (.pkl or .joblib)
- Feature schema file (feature_schema.json)

Phase 4 — Model Evaluation and Explainability

Business Goal

Ensure that predictions are reliable, interpretable, and safe for hospital operations and finance teams.

- Evaluate both models on training and test data using confusion matrices and precision, recall, and F1-score.
- Compute business metrics such as recall for High Risk class and recall for Rejected claim class.
- Generate feature importance and explainability summaries.
- Segment performance by gender, city, and insurance_provider to identify fairness gaps.
- Create a consolidated model card documenting performance, limitations, and assumptions.

Deliverables

- Risk model evaluation report
- Claim model evaluation report
- Model card document
- Explainability summary

Recommendation (To handle imbalance data)

1. **Tune Hyperparameters:** Adjust class weights or tree depth.
2. **Feature Engineering:** Create more interaction features (e.g.,
 - bill_amount
 - vs
 - department
 - averages).
3. **Resampling:** Use SMOTE or undersampling to handle the class imbalance more aggressively.

Phase 5 — Deployment and API Integration

Business Goal

Make models accessible to hospital systems and dashboards in real time.

- Build a FastAPI service with endpoints for health checks and predictions.
- Implement request and response schema validation.
- Log predictions with timestamp, model version, and input feature hash.
- Containerize the service using Docker.
- Prepare a deployment and operations runbook.

Deliverables

- API source code
- Dockerfile
- Deployment guide
- Sample request and response documentation

Phase 6 — Monitoring, Drift Detection, and Governance

Business Goal

Ensure long-term reliability, safety, and compliance of the AI system.

- Implement data validation checks for missing values, numeric ranges, and unseen categories.
- Track feature and prediction drift over time.
- Maintain an audit log for predictions and model versions.
- Document system limitations, assumptions, and retraining strategy.

Deliverables

- Monitoring scripts
- Drift detection report
- Governance and compliance document

Final Phase — Executive Business Presentation

Objective

Translate technical outcomes into leadership-level business value.

- Business problem and operational risks
- System architecture and data flow
- Key insights from SQL and EDA
- Model performance in business terms
- Financial impact and ROI estimation
- Deployment and scaling strategy
- Risk management and compliance plan